Compositional Taylor expansion

(in cartesian differential categories)

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This talk

Differential calculus + Compositionality

Why differential calculus and compositionality matters

- Automated differentiation

Why differential calculus and compositionality matters

- Automated differentiation
- Differential λ-calculus and differential Linear Logics

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- **Categorical differentiation**

Why differential calculus and compositionality matters

- Automated differentiation
- Differential λ-calculus and differential Linear Logics
- **Categorical differentiation**

Categorical differentiation

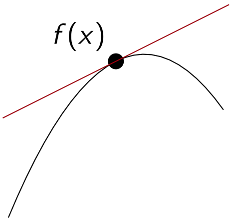
- Axiomatization of derivatives in categories
- At first: axiomatization of models of differential λ-calculus

Let us go back to undergraduate class on differential calculus

Differential calculus

$f : \mathbb{R} \to \mathbb{R}$ derivable at $x \in \mathbb{R}$

$$f(x+u) = f(x) + f'(x) \cdot u + O(u^2)$$



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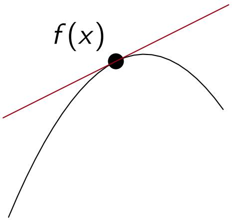
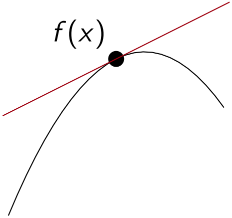


Derivative $\begin{aligned} f' : \mathbb{R} &\to \mathbb{R} \\ x &\mapsto f'(x) \end{aligned}$

Differential calculus

$f : E \to F$ derivable at $x \in E$

$$f(x+u) = f(x) + f'(x) \cdot u + O(\|u\|^2)$$



$$\text{Derivative} \quad \begin{array}{rcl} f' : E & \to & E \multimap F \\ x & \mapsto & u \mapsto f'(x) \cdot u \end{array}$$

Differential calculus

$$\text{Derivative} \quad f' : E \rightarrow E \multimap F$$
$$x \mapsto u \mapsto f'(x) \cdot u$$

Differential calculus

$$\text{Derivative} \quad \begin{aligned} f' : E &\to E \multimap F \\ x &\mapsto u \mapsto f'(x) \cdot u \end{aligned}$$

$$\text{Order 2 Derivative} \quad \begin{aligned} f^{(2)} : E &\to E \multimap E \multimap F \\ x &\mapsto (u_1, u_2) \mapsto f^{(2)}(x) \cdot (u_1, u_2) \end{aligned}$$

Differential calculus

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Degree 2 Taylor expansion

$$\boxed{f(x+u) = f(x) + f'(x) \cdot u + \frac{1}{2} f^{(2)}(x) \cdot u^2 + O(\|u\|^3)}$$

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Degree n Taylor expansion.

$$\boxed{f(x+u) = \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(x) \cdot (\underbrace{u, \ldots, u}_{k \text{ times}}) + O(\|u\|^{n+1})}$$

First step: How do we make derivative compositional ?

Explicit category theory: only at the end.

Fixing the chain rule

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x)$$ ☹

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Tangent bundle

$$\mathrm{T}X = X \times X \qquad \frac{f : X \to Y}{\mathrm{T}f : \mathrm{T}X \to \mathrm{T}Y} \qquad \mathrm{T}f(x, u) = (f(x), f'(x) \cdot u)$$

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Dual number intuition

$$(x, u) \in \mathrm{T}X \iff \text{polynomial } x + u\varepsilon \text{ with } \varepsilon^2 = 0$$

$$f(x + u\varepsilon) = f(x) + \varepsilon f'(x) \cdot u \ + O(\varepsilon^2)$$

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$$f(x + u\varepsilon) = f(x) + \varepsilon f'(x) \cdot u \ + \cancel{O(\varepsilon^2)}$$ **push forward**

Fixing the higher order chain rule

Faá di Bruno formula: bad (and terrifying) ☹

$$\cancel{(g \circ f)^{(n)}(x) = \sum_{\{I_1,\dots,I_k\} \text{ partition of } [\![1,n]\!]} g^{(k)}\left(f(x), f^{(I_1)}(x), \ \dots \ , f^{(I_k)}(x)\right)}$$

with $f^{(\{i_1,\dots,i_l\})}(x) : (u_1, \dots, u_n) \mapsto f^{(l)}(x)(u_{i_1}, \dots, u_{i_l})$

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$$\mathrm{T}^n(g \circ f) = \mathrm{T}^n g \circ \mathrm{T}^n f$$ ☺

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$$a \in \mathrm{T}^n X = X^{2^n} \iff a(\varepsilon_1, \cdots, \varepsilon_n) \text{ polynomial with } \varepsilon_i^2 = 0$$

$\mathrm{T}^n f : \mathrm{T}^n X \to \mathrm{T}^n Y$: **pushes forward** polynomials

$\mathrm{T}f(x, u) = (f(x), f'(x) \cdot u)$: order 1 Taylor expansion

Contribution: we do the same for Taylor expansion

A compositional Taylor expansion

Tangent bundle

$$\mathsf{T}X = X \times X \qquad\qquad \frac{f : X \to Y}{\mathsf{T}f : \mathsf{T}X \to \mathsf{T}Y}$$

A compositional Taylor expansion

Taylor bundle

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Higher order dual numbers intuition

$$(x, u, v) \in \mathrm{T}_2 X \iff \text{polynomial } x + u\varepsilon + v\varepsilon^2 \text{ with } \varepsilon^3 = 0$$

$\mathrm{T}_2 f(x, u, v)$: **push forward** of polynomial.

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$$\begin{aligned} & f(x + u\varepsilon + v\varepsilon^2) \\ & = f(x) + f'(x) \cdot (u\varepsilon + v\varepsilon^2) + \frac{1}{2} f^{(2)}(x) \cdot (u\varepsilon + v\varepsilon^2, u\varepsilon + v\varepsilon^2) + \cancel{O(\varepsilon^3)} \end{aligned}$$

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$$\begin{aligned}
& f(x + u\varepsilon + v\varepsilon^2) \\
&= f(x) + f'(x) \cdot (u\varepsilon + v\varepsilon^2) + \frac{1}{2} f^{(2)}(x) \cdot (u\varepsilon + v\varepsilon^2, u\varepsilon + v\varepsilon^2) + \cancel{O(\varepsilon^3)} \\
&= f(x) + \varepsilon f'(x) \cdot u + \varepsilon^2 \left(\frac{1}{2} f^{(2)}(x) \cdot (u, u) + f'(x) \cdot v \right) + \cancel{\varepsilon^3(\ldots)}
\end{aligned}$$

A compositional Taylor expansion

$$\mathrm{T}_n f(x, u_1, \ldots, u_n) = \left(\sum_{k=1}^{j} \sum_{\substack{i_1,\ldots,i_k \text{ s.t.} \\ i_1+\cdots+i_k=j}} \frac{1}{k!} f^{(k)}(x) \cdot (u_{i_1}, \ldots, u_{i_k}) \right)_{j=0}^{n} \qquad (1)$$

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Generalizes Taylor expansion

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Is compositional

$$\mathrm{T}_n(g \circ f) = \mathrm{T}_n g \circ \mathrm{T}_n f$$

Proof: by... equation 1 ? ☹

A simpler proof by abstract nonsense

$\mathrm{T}_n f$ and $\mathrm{T}^n f$ are **push forward**.

- $\mathrm{T}^n X = X^{2^n}$: polynomial $a(\varepsilon_1, \ldots, \varepsilon_n)$, with $\varepsilon_i^2 = 0$
- $\mathrm{T}_n X = X \times X^n$: polynomial $a(\varepsilon)$, with $\varepsilon^{n+1} = 0$

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A natural transformation

injection $\mathsf{L}_n : \mathsf{T}_n X \to \mathsf{T}^n X$

$$\mathsf{T}^n f \circ \mathsf{L}_n = \mathsf{L}_n \circ \mathsf{T}_n f \quad ☺$$

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$$\mathrm{L}_n \circ \mathrm{T}_n(g \circ f) = \mathrm{T}^n(g \circ f) \circ \mathrm{L}_n = \mathrm{T}^n g \circ \mathrm{T}^n f \circ \mathrm{L}_n = \mathrm{L}_n \circ \mathrm{T}_n g \circ \mathrm{T}_n f$$

Category theory

Compute on Taylor expansions without writing them down

Categorical structure of T

T exists in all **cartesian differential categories**.

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T is a functor	$\iff$	Chain rule
T is a monad	$\iff$	$u \mapsto f'(x) \cdot u$ linear
Distributive law $c : TT \Rightarrow TT$	$\iff$	Symmetry of 2nd order derivative

Same categorical structure on T_n

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- T_n is a functor (Done)
- T_n is a monad (Todo)
- $\mathrm{c}_{n,m} : \mathrm{T}_n\mathrm{T}_m \Rightarrow \mathrm{T}_m\mathrm{T}_n$ is a natural transformation (Todo)

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Let us compute... $\mathrm{T}_n^2 f$?

Proof by... computation ?

Remember

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Let us compute... $\mathrm{T}_n^2 f$? ☹

Let us use categories! ☺

Category saves the day

- T **is a monad**
- T^n is a monad
- $\mathrm{L}_n : \mathrm{T}_n X \to \mathrm{T}^n X$ strikes again

Easy

Category saves the day

- T is a monad
- T^n **is a monad**
- $\mathsf{L}_n : \mathsf{T}_n X \to \mathsf{T}^n X$ strikes again

Category theory tools
No computations, all induction

Distributive law $\mathsf{c} : \mathsf{TT} \Rightarrow \mathsf{TT}$
and Yang-Baxter equation

Category saves the day

- T is a monad
- T^n is a monad
- $\mathrm{L}_n : \mathrm{T}_n X \to \mathrm{T}^n X$ **strikes again**

Same as proof of $\mathrm{T}_n(g \circ f) = \mathrm{T}_n g \circ \mathrm{T}_n f$

$\mathrm{L}_n : \mathrm{T}_n X \to \mathrm{T}^n X$ is a **monad transformer**

Category theory concepts captures Taylor expansion

Differential calculus	Category theory
Faá di Bruno	Functoriality of T_n
Dual numbers	Kleisli category of monad T_n
Schwarz theorem	Distributive law $\mathrm{c} : \mathrm{T}_n\mathrm{T}_n \Rightarrow \mathrm{T}_n\mathrm{T}_n$
Partial derivatives	Monad strength
Multiple dual numbers $\varepsilon_1, \ldots, \varepsilon_l$	Canonical monad $\mathrm{T}_{n_1} \cdots \mathrm{T}_{n_l}$

All proof on the category theory world: <u>no combinatorics</u>

We can also do

infinitary Taylor expansions

To conclude

Compositional approach to

- derivatives T
- higher derivatives T^n
- degree n Taylor expansion T_n
- infinitary Taylor expansion T_ω

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No combinatorics on Taylor series

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Compositional approach to

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No combinatorics on Taylor series

- Monad transformer $\mathrm{L}_n : \mathrm{T}_n X \to \mathrm{T}^n X$.
- Categories (Monad, strength, distributive law) $\iff$ properties of differential calculus

Future work

- Connection with automated differentiation for higher order derivatives
- Connection with jet bundles (tangent categories)
- What about reverse mode Taylor expansion ?